

Circulatory Physiology

1- The function of the ... is to transport blood under high pressure to the tissues.

- ✓ arteries
- veins
- arterioles
- venules

2- The ... collect blood from the capillaries and gradually coalesce into progressively larger veins.

- ✓ venules
- arteries
- veins
- arterioles

3- The ... in the capillaries tends to force fluid and its dissolved substances through the capillary pores into the interstitial spaces.

- Osmotic pressure
- Colloid pressure
- Lymphatic pressure
- ✓ Hydrostatic pressure

4- The approximate average forces of net filtration pressure of the capillary is about

- 20 mmHg
- 2 mmHg
- 1 mmHg
- ✓ 13 mmHg

5- The ... are the last small branches of the arterial system; they act as control conduits.

- ✓ arterioles
- venules
- veins
- arteries

6- Whenever people with varicose veins stand for more than a few minutes,

Edema in leg might detect.



They faint.

Albumin level increase.

Coronary arteries collapse.

7- The overall blood flow in the total circulation of an adult person at rest is about:

1000 ml/min

120 ml/min



5000 ml/min

2000 ml/min

8- Which molecule has less permeability?



Albumine

Glucose

Urea

Water

9- Increasing ... directly increases blood viscosity.



hematocrite

muscles

serum plasma volume

body weight

10- In the, the junctions between the capillary endothelial cells are mainly tight junctions.

kidney

spleen

liver



brain

11- Which one is not considered as specific blood reservoirs?

Spleen



Aorta

Liver

Large abdominal veins

12- The rate of blood flow through the entire circulatory system is equal to the

rate of blood pumping by the lung



cardiac output

Pulmonary artery

Brain artery

13- Inhibition of ... greatly dilates the vessels and can increase the blood flow.



sympathetic activity

parasympathetic activity

GH

liver activity

14- In persons with aortic valve stenosis, the aortic pressure pulse is ... significantly.

increase

no change



decrease

opened

15- Which vessels are much more distensible than the arteries?

Arterioles

Heart

Blood



Veins

16- Which one has important role on colloid osmotic pressure?

Inulin

Hydrogen ion

✓ Albumine

Water

17- Aortic arc baroreceptors send signaling to the vasomotor center by ... nerve.

✓ Vagus

Glossopharyngeal

Carotid

Hering's

18- In which situation, hematocrite is higher than normal range?

Anemia

None of them

Low red blood cell count

✓ Polycythemia

19- Which part of brain controls vasoconstriction?

Amygdala

Hippocampus

Cerebellum

✓ Vasomotor center

20- Which factor could decrease blood viscosity?

Erythropoietin enhancement

Polycythemia

Smoking

✓ Anemia

21- Based on ohm's law, if pressure difference increase, blood flow will be

- safe
- decrease
- not change
- ✓ increase

22- Where is the position of peripheral baroreceptors?

- ✓ Carotid sinus and aortic arc
- Lung
- Venuls
- Cerebrum

23- Which hormone act as vasoconstrictor?

- LH
- ✓ Vasopressin
- Glucagon
- TSH

24- Which one is not seen during muscular activity (exercise)?

- Vasoconstriction in kidney
- ✓ Vasodilatation in gastrointestinal vessels.
- Vasodilatation in related muscles
- Increase in heart rate

25- Parasympathetic stimulation ... heart rate and contractility.

- ✓ decreases
- increases
- has no role on
- indirectly stimulate

26- The first cell that can be identified as belonging to the RBC series is

reticulocyte

CFU-B

CFU-S

✓ proerythroblast

27- Opsonization means

✓ a process by which a pathogen is selected for phagocytosis and destruction

a process by which the tissue spaces and the lymphatics in the inflamed area are blocked by fibrinogen clots

a process by which neutrophils and monocytes squeeze through the pores of the blood capillaries

a process by which white blood cells are attracted to inflamed tissue areas

28- The least amount of iron in human body is found in the form of

ferritin

✓ transferrin

myoglobin

hemoglobin

29- The first effect of leukemia is

✓ metastatic growth of leukemic cells in abnormal areas of the body

very few WBCs production in bone marrow

severe anemia, and a bleeding tendency caused by thrombocytopenia

causing pain and, eventually, a tendency for bones to fracture easily

30- Thrombin does not have a direct effect on

factor VIII

✓ factor VI

factor V

prothrombin

31- Which of the following is not considered to be an intravascular anticoagulants?

The smoothness of the endothelial cell surface



Plasmin

Heparin

The fibrin fibers that are formed during the process of clotting

32- Which of the following refers to the last line of defense against infection?

Neutrophil invasion into the inflamed area



Increased production of granulocytes and monocytes by the bone marrow

Acute increase in the number of neutrophils in the blood

Second macrophage invasion into the inflamed tissue

33- Which of the following refers to the main factor in the control mechanism of erythropoietin secretion?

Central nervous system stimulation

Concentration of RBCs in the blood



Tissue oxygenation

High blood volume

34- After blood vessels rupture, which of the following initiates the extrinsic pathway?

Factor IX



Factor III

Factor XII

Factor XI

35- Which type of maturation failure of RBCs is caused as a result of intrinsic factor deficiency?

Hereditary spherocytosis



Megaloblastic anemia

Hemolytic anemia

Aplastic anemia

36- Each heme molecule is formed from

4 pyrrole

✓ protoporphyrin IX + Fe^{++}

polypeptide + Fe^{++}

2 α chains + 2 β chains

37- Disseminated intravascular coagulation often results from

loss of the von Willebrand factor

✓ the presence of large amounts of traumatized or dying tissue in the body

an abnormality or deficiency of Factor VIII

the presence of very low numbers of platelets in the circulating blood

Heart Physiology

38- The QT interval in the electrocardiogram trace represents

ventricular refractory period

ventricular depolarization

✓ ventricular contraction

ventricular tension time index

39- A decrease in atrioventricular nodal conduction velocity will

decrease heart rate

widen the QRS complex

✓ increase PR interval

increase P wave amplitude

40- Ventricular end-diastolic pressure can be used instead of the usual variable on the abscissa (x-axis) when plotting a Frank-Starling curve. What variable does the x-axis usually represent on a Frank-Starling curve?

cardiac work

cardiac output

✓ ventricular end-diastolic volume

ventricular stroke volume

41- In slow response action potential cells of the cardiac myocytes, phase 0 is mainly achieved by

Fast Na^+ channels

K_1 inwardly rectified channels

✓ L- type Ca^{++} channels

Na funny channels

42- The transmembrane potential during phase 4 is much less negative in nodal cells than in atrial and ventricular myocytes because

✓ nodal cells lack the i_{K1} type of K^+ channel

nodal cells lack the funny Na^+ channel

nodal cells have high i_{K1} type of K^+ channel

nodal cells have high funny Na^+ channel

43- During which phase of the cardiac cycle does the mitral valve open?

atrial systole

reduced ventricular filling

isovolumetric ventricular contraction

✓ isovolumetric ventricular relaxation

44- In a standard lead III electrocardiography, the negative terminal is connected to

the right leg

✓ the left arm

the right arm

the left leg

45- The ionic basis of the automaticity in the nodal cells that mediates the slow diastolic depolarization is all but

an inward Ca^{++} current through L-type Ca^{++} channels (i_{Ca})

✓ an inward rectifying K^+ current (K_1)

hyperpolarization-induced Na^+ inward current (i_f)

an outward i_{K} current (i_{K})

46- The duration of the plateau in atria is substantially less than in ventricular myocytes because

none of the above

the magnitude of i_{to} during plateau is greater in atria than in ventricular myocytes



the magnitude of i_{to} during plateau is less in atria than the ventricular myocytes

the magnitude of i_{to} during plateau is the same in atria and ventricular myocytes

47- The time from the upstroke of the carotid artery pulse to the incisura is a measure of the period of

atrial diastole



ventricular ejection

reduced ventricular filling

rapid ventricular filling

48- The s wave in an electrocardiogram represents



depolarization of a small portion of the base of left ventricle

depolarization of the septum from left to right

depolarization of the apex

repolarization of a small portion of the base of the left ventricle

49- An increase in systemic blood pressure leads to which of the following effects?

an increase in the velocity at which blood ejected from left ventricle

a decrease in the maximal wall tension developed in the left ventricular muscle

an increase in cardiac output



an increase in residual volume of blood in the left ventricle

50- A hospitalized patient has an ejection fraction of 0.4, a heart rate of 95 beats/min, and cardiac output of 3.5 L/min. what is the patient's end-diastolic volume?

14 ml



92 ml

55 ml

37 ml

51- If the R wave is upright and equally large on leads I and aVF, what is the mean electrical axis of the heart?

0 degree



45 degrees

135 degrees

90 degrees

52- Administration of certain K channel antagonists causes the plateau of an action potential curve to be

doesn't affect phase 2 AP phase duration

unchanged



prolonged

diminished

53- Very high sympathetic neural activity to the heart can lead to all of the following effects except

positive dromotropism

positive chronotropism

positive bathmotropism



cardiac tetanic contractions

54- The T wave in a normal ECG trace is positive because

the apex of the left ventricle mass is the last to repolarize

the inner surfaces of the ventricles repolarize sooner than the outer apical surfaces

the septum and endocardial areas of the ventricular muscle depolarizes first



the septum and endocardial areas have longer period of contraction than do most of the heart external surface

55- The v wave in the atrial pressure curve that occurs toward the end of ventricular contraction is due to

slight backflow of blood into the atria at the onset of ventricular contraction

atrial contraction

bulging of the A-V valves backward toward atria



slow flow of blood into atria from veins while A-V valves are closed

56- The most common causes of decreased QRS complex are all but

- ✓ ventricular hypertrophy
- pleural fluid accumulation (effusion)
- cardiomyopathy
- pericardial fluid retention (pericardial effusion)

57- High tension-time index in the cardiac muscle cells occur during

- isovolumetric contraction
- ✓ abnormally dilated left ventricle
- isovolumetric relaxation
- normal left ventricular diameter

58- The P wave on lead aVR of the normal electrocardiogram will be

- ✓ a downward deflection
- highly variable
- not detectable
- an upward deflection

Anatomy of cardiovascular block

59- All of the following structures are visible posterior to thoracic aorta except

- Thoracic vertebra
- Accessory hemiazygous vein
- Hemi Azygous vein
- ✓ Right posterior intercostal artery

60- All of the following structures are seen in both the upper mediastinum and the posterior mediastinum, except ...?

- esophagus
- Right Vagus nerve
- ✓ Right recurrent laryngeal nerve
- Left Vagus nerve

61- Which of the following structures located at the lower end of the vertebral canal?

coccyx

Sacral crest

Iliac crest



Sacral hiatus

62- Which of the following structure is found both inferior and posterior the arch of the aorta?

Left brachiocephalic vein

Left phrenic



Left recurrent laryngeal nerve

Left vagus nerve

63- Where is sinu-atrial node?

In left atrium

In right ventricle

In left ventricle



In right atrium

64- All the following options are correct except ...?

The pericardio co phrenic vessels accompany the phrenic nerve in the chest

The thoracic aorta begins on the left side of the fourth thoracic vertebra



The thymus is found in the posterior mediastinum

The left brachiocephalic vein is longer than the right

65- In the case of the sympathetic chain, all of the following definitions are correct except?

The sympathetic chain is also seen in the abdomen

White rami is find in lateral side to gray rami



White rami is along the entire length of the sympathetic chain

Greater splanchnic nerves contain myelin

66- During a heart procedure the surgeon accidentally damages the vein that is accompanied by the posterior interventricular artery, which of the following veins is most likely to be damaged?

Great cardiac vein

Small cardiac vein

✓ Middle cardiac vein

Left marginal vein

67- Where we should listen to Mitral valve sounds?

In 2th left intercostal space

✓ In left 5th intercostal space over apex of heart

In 6th right intercostal space

In 2th right intercostal space

68- All of the following arteries arises from internal thoracic artery except...?

Sternal artery

Musculophrenic artery

✓ Posterior intercostal artery

Perforating artery

69- Where is the deep cardiac plexus has seen?

✓ Anterior to bifurcation of trachea

In anterior mediastinum

In posterior mediastinum

Inferior to arch of Aorta

70-

Which of the following muscles is in the deep layer of the erector spin muscle?

Multifidus

✓ Rotatores

Longissimus thoracis

Semispinalis

71- All of the following structure are find in right ventricle except...

Septomarginal

Papillary muscle

Chorda tendino

✓ Apex of heart

72- All of following ribs has a single articular facet on its head except?

first rib

✓ Third rib

Twelve rib

tenth rib

73- 9th left posterior intercostal vein is going to drain into

Inferior vena cava

Superior vena cava

Azygous vein

✓ Hemiazygous vein

74- Lymphatic drainage of all the following areas into the thoracic duct except ..?

Left upper limb

Left lower limb

Right lower limb

✓ Right upper limb

75- -All of the following structures are going to innervate by phrenic nerve except which?

✓ Visceral serous pericard

Diaphragm

Parital serous preicard

Fibros pericard

76- Which of the following structures crosses the anterior arch of the aorta

Right vegus nerve



Left vagus nerve

Left recurrent laryngeal nerve

Right phrenic nerve

77- Which rib is longest rib in human body?

5th

6th



7th

4th

78- The right margin of heart is formed mostly by which champer of the heart?

Left atrium

Left ventricle



Right atrium

Right ventricl

79- Which of the opening of following vein in heart has valve..?



Coronary sinus

Superior vena cava

Right pulmonary vein

Left pulmonary vein

80- The upper two posterior intercostal arteries on each side are derived which artery



Subclavian

Thoracic aorta

Right internal thoracic

Left Internal thoracic

81- Which branches of intercostal nerve is largest?

- ☐ Anterior terminal branch
- ☒ Lateral cutaneous branch
- ☐ Collateral branch
- ☐ Posterior branch

82- All of the following arteries arises from left coronary in right dominant heart except.....

- ☐ left marginal
- ☐ Circumflex
- ☒ Posterior interventricular branch
- ☐ Diagonal

83- Which chamber of the heart in chest has anterior position?

- ☐ Right atrium
- ☒ Right ventricle
- ☐ Left ventricle
- ☐ Left atrium

Histology

84- Which antibody has pentamer structure?

- ☐ Ig E
- ☒ Ig M
- ☐ Ig G
- ☐ Ig A

85- Which organ has dual embryonic origin?

- ☐ Spleen
- ☐ Lymphnode
- ☒ Thymus
- ☐ Lingual tonsil

86- Most numerous vessels in our body are

- ☐ Large arteries
- ☒ capillaries
- ☐ Small arteries
- ☐ veins

87- Where is the site of blood filtration in body?

- ☐ tonsil
- ☒ spleen
- ☐ thymus
- ☐ lymphnode

88- Which one is the antibody of secretions such as saliva?

- ☐ Ig E
- ☐ Ig M
- ☐ Ig G
- ☒ Ig A

89- The thickest layer in the wall of veins are....

- ☐ subendothelia
- ☐ media
- ☐ intima
- ☒ adventitia

90- Which one is primary lymphoid organ?

- ☐ Tonsil
- ☐ Lymphnode
- ☐ Spleen
- ☒ Thymus

91- Where we observe Hassall's corpuscle in thymus?

- ☐ capsule
- ☒ Medulla of lobule
- ☐ Cortex of lobule
- ☐ trabecula

92- Which one is the largest lymphatic organ in the body?

- ☐ thymus
- ☐ Lymph node
- ☒ Spleen
- ☐ Palatine tonsil

93- In which vessel we observe internal elastic layer obviously?

- ☒ Muscular artery
- ☐ capillary
- ☐ Elastic artery
- ☐ Medium sized veins

Embryology

94- Which of the following septum forms the floor of the limbus fossa ovalis?

- ☐ septum primum.
- ☐ sub endocardial cushions.
- ☒ septum secundum.
- ☐ septum intermedium

95- During vascular development, the hepatic part of IVC arises from:

- ☒ Right vitelline veins.
- ☐ Right subcardinal veins.
- ☐ Right supracardinal veins
- ☐ Right umbilical veins.

96- Which part of the primitive heart tube gives rise to the right ventricle and the aorta?

Primitive ventricle

Sinus venosus

✓ Bulbus cordis

Primitive atrium

97- During Aortic arch development, Fate of the following are correct EXCEPT:

✓ Aortic sac: distal (descending) segment of the aortic arch and brachiocephalic trunk.

1st aortic arch: maxillary artery (in the face)

3rd aortic arch: common carotid and initial segments of internal carotid artery

2nd aortic arch: stapedial artery (in middle ear)

98- Which of the follow is NOT part of TETRALOGY OF FALLOT:

✓ ASD

Thicked (Hypertrophy)right ventricle wall

Overriding of the aorta

Pulmonary stenosis

99- The right horn of sinus venous forms:

The Rough Anterior wall of the right atrium.

The oblique vein

✓ The Smooth Posterior wall of the right atrium.

Atrophy and forms coronary sinus

100-Failure of the aorticopulmonary septum to undergo the 180-degree spiral will result in

An atrial-septal defect

Persistent truncus arteriosus

Tetralogy of Fallot

✓ Transposition of the great vessels